Seaborn - CATPLOT METHOD

📘 sns.catplot()

🔹 **catplot()**

- **Type:** Function
- **Use:** Creates **categorical plots** with support for **multiple subplots (faceting)**
- **Returns:** FacetGrid object

🔹 **data**

- **Type:** DataFrame
- **Use:** Dataset from which values are taken

data = tips

🔹 **x**

- **Type:** column name / array
- **Use:** Categorical variable (plotted on X-axis)

x = "day"

🔹 **y**

- **Type:** column name / array
- **Use:** Numerical variable (plotted on Y-axis)

y = "total_bill"

🔹 **hue**

- **Type:** column name
- **Use:** Adds **color grouping (subcategory)**

hue = "sex"

👉 Output: Different colors for Male/Female

🔹 **kind**

- **Type:** string
- **Use:** Defines **type of plot**

Possible values:

- "strip" (default)
- "swarm"
- "box"
- "violin"
- "boxen"
- "point"
- "bar"
- "count"

kind = "bar"

🔹 **row**

- **Type:** column name
- **Use:** Creates **multiple rows of plots**

row = "smoker"

👉 Separate plots for smoker = Yes / No

🔹 **col**

- **Type:** column name
- **Use:** Creates **multiple columns of plots**

col = "time"

👉 Separate plots for Lunch / Dinner

🔹 **col_wrap**

- **Type:** integer
- **Use:** Wraps columns into multiple rows

col_wrap = 2

👉 After 2 columns, next plot moves to next row

🔹 **legend**

- **Type:** True / False
- **Use:** Show or hide legend

legend = True

🔹 **estimator**

- **Type:** function
- **Default:** mean
- **Use:** Defines how values are aggregated

Common values:

- np.mean
- np.median

- np.sum
- custom function

```
import numpy as np
estimator = np.sum
```

👉 Example:

- mean → average
- sum → total
- median → middle value

🔹 **Combined Example**

```
import seaborn as sns
import numpy as np

tips = sns.load_dataset("tips")

sns.catplot(
  data=tips,
  x="day",
  y="total_bill",
  hue="sex",
  kind="bar",
  row="smoker",
  col="time",
  col_wrap=2,
  estimator=np.mean,
  legend=True
```